HAMZA AZIZ

AI Engineer | Voice AI, Generative AI & Computer Vision

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Nationality: Indian | Notice period: 30 days

Open to relocation: Abu Dhabi (preferred) and Dubai

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PROFESSIONAL SUMMARY

AI Engineer with 3+ years shipping production voice AI, LLM and computer-vision systems. Helped lift live drive-thru ordering success from ~25–30% to 60%+ on a multi-tenant Gemini platform across 4 POS integrations, and built, as sole developer, face recognition across 20+ live CCTV streams at >99.99% accuracy in controlled UAT. Seeking applied AI roles in Abu Dhabi or Dubai.

WORK EXPERIENCE

AI Product Engineer | *VoicePlug Inc.* | September 2025 – Present

- Helped raise live ordering success from ~25–30% to 60%+ by fixing intent, entity, menu-understanding and fallback failures found in 500+ real customer calls
- Run a production FastAPI + Socket.IO voice-ordering platform at 2 live restaurants (4+ QSR brands), with Redis session state, a MongoDB order book and a Spanish-language ordering mode
- Bounded LLM cost and latency with per-store Vertex AI context caching, Flash/Flash-Lite tiering across 5 model roles and a 1,024-token reasoning cap
- Kept pricing and checkout out of the model: one cached Gemini 2.5 Flash call with no tool calls, inside deterministic guards, integrated with 4 POS systems (QUBeyond, Clover, Speedline, Deliverect)
- Eliminated cart crashes with a Redis-locked per-session cart worker, combo and size guards, and fuzzy matching for speech-recognition disfluencies
- Built a replay test harness (each case run 3× for model variability) and a review loop from production logs to fixes

Assistant Engineer – AI | *Global InfoVentures Pvt. Ltd.* | August 2023 – August 2025

- Designed and built the CCTV face-recognition attendance application as sole developer on InsightFace-REST (SCRFD, w600k_r50), TensorRT and Milvus, reaching >99.99% accuracy against the attendance register in controlled UAT; delivered 4 CCTV, PTZ and edge variants across DGX A100, RTX 4090 and Jetson Orin Nano
- Processed 20+ live CCTV streams with ROI masking and low-light gating, batching 32 frames across 4 TensorRT recognition threads on A100 MIG slices, with Kafka events feeding Raspberry Pi displays
- Lifted real-world recognition accuracy from 85% to 99% with nightly staging-gallery updates from elevated-angle footage; combined mobile and CCTV datasets improved cross-domain recognition by 14%
- Built FastAPI reporting on partitioned PostgreSQL with 1-minute ERP sync for a 20-camera main gate averaging ~2,000 people a day
- Built and deployed a PTZ classroom-attendance system as sole developer: ONVIF preset sweeps, InsightFace-REST with Milvus matching, YOLOv9 headcounts on a GPU MIG slice and ERP exports every 5 minutes
- Developed G6, a Dockerised FastAPI voice ERP assistant for 4 institutes, with two-stage Rasa routing for student, employee and fee queries
- Grounded Gemini in a per-institute Milvus index of schema fields, aliases and few-shot examples, so it emits fields and conditions for application-built ERP queries rather than full SQL

AI Intern | *Global InfoVentures Pvt. Ltd.* | February 2023 – July 2023

- Delivered 10+ AI/ML training sessions, mentored 50+ students across CSE, AIML and Data Science, and guided 10+ YOLOv8/CNN object-detection projects through deployment; built a browser-based PPE violation monitor using a custom YOLOv7 model, Flask, OpenCV and PyTorch, with a 0.75 confidence gate and Gmail API email alerts with cooldown control

PROJECTS

- **Jetson Nano Face Recognition Attendance (offline, on-device):** DeepStream pipeline with SCRFD + ArcFace on TensorRT FP16; moving detection into an adaptive pad probe cut idle GPU load from 89–99% to near zero and raised throughput from 12.2 to 15.0 fps on a 1080p camera, with self-installing releases and automatic rollback
- **Dubai Smart Irrigation Advisor:** ranks 6 irrigation predictors by operating cost in AED; its FAO-56 physics core matches Dubai climate normals to 2.3% over 30 years of NASA POWER data, and acting on raw soil sensors alone costs 6.7× more
- **Clinic Voice Receptionist:** voice appointment booking in 5 languages (English, Hindi, Tamil, Kannada, Malayalam) at 0.8–1.0 s per turn, with no unflagged wrong detail even at 50% simulated transcription error
- **Codex–Claude Council:** npm plugin and MCP server listed in the MCP Registry, Smithery and awesome-mcp-servers; scored 24/24 on a tricky-program benchmark

SKILLS

- **Generative & Conversational AI:** Google Gemini, Vertex AI, Context Caching, LangGraph, LangChain, Agentic AI, RAG, LLMs, vLLM, Prompt Engineering, Rasa, NLP
- **Machine Learning & Computer Vision:** PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, InsightFace, NumPy, Pandas
- **Inference & Edge AI:** TensorRT, ONNX, DeepStream, NVIDIA Jetson, DGX A100
- **Programming & Backend:** Python, SQL, Bash, TypeScript, FastAPI, Socket.IO
- **Data, Messaging & DevOps:** PostgreSQL, MongoDB, Redis, Milvus (Vector DB), Kafka, MinIO, Docker, CI/CD, Git, Linux

LANGUAGES

- English (Professional Working Proficiency)
- Hindi (Native)

EDUCATION

Master of Technology (Executive), Artificial Intelligence and Machine Learning
BITS Pilani — Work Integrated Learning Programmes (WILP) | 2025 – 2027 (in progress)

Bachelor of Technology, Computer Science and Engineering
ABES Institute of Technology, Dr. A. P. J. Abdul Kalam Technical University (AKTU) | 2019 – 2023

CERTIFICATES

- Rasa Developer Certification — 2025
- Building Video AI Applications at the Edge on Jetson Nano, NVIDIA Deep Learning Institute — 2024
- Getting Started with Deep Learning, NVIDIA Deep Learning Institute — 2022